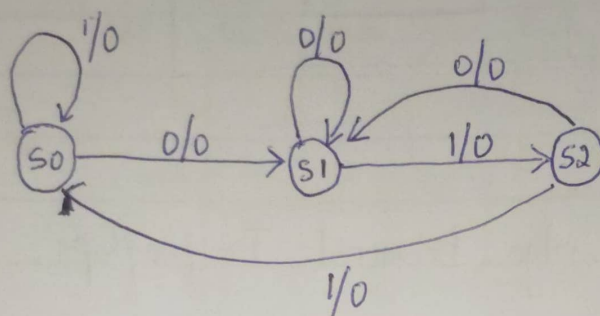


Day-9

FSMs - Assignment

- 1) Draw the state diagram for a FSM, the output toggles whenever it detects '110' LSB first (overlapping). Consider initial value of output as 0.

⇒ State Diagram:



Previous state		input I	next state		output (Y)
q_1	q_0		D_1	D_0	
S_0	0	0	0	1	0
	0	1	0	0	0
S_1	0	0	0	1	0
	0	1	1	0	0
S_2	1	0	0	1	0
	1	0	0	0	1
X	X	X	X	X	0
X	X	X	X	X	0

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$$D_1 =$$

$q_1 \backslash q_0 \bar{I}$	00	01	11	10
0	0	0	1	0
1	0	0	X	X

$$D_0 =$$

$q_1 \backslash q_0 \bar{I}$	00	01	11	10
0	1	0	0	1
1	1	0	X	X

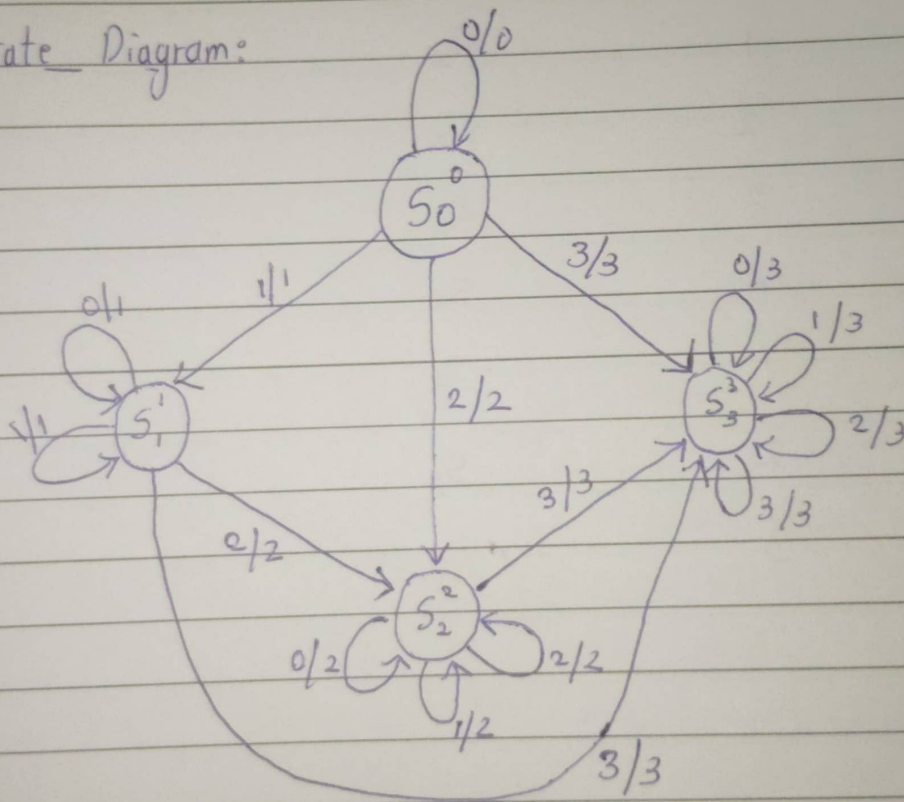
$$\therefore D_1 = q_0 \bar{I}$$

$$D_0 = \bar{I}$$

$$Y = q_1 \bar{q}_0 \bar{I}$$

- 2) Design a FSM over $\{0, 1, 2, 3\}$ that will output the largest input digit read so far. e.g. input: 001032 will produce output: 001133.

$\Rightarrow$ State Diagram:



Previous state	Input	next state
S_0	0	S_0
S_0	1	S_1
S_0	2	S_2
S_0	3	S_3
S_1	0	S_1
S_1	1	S_1
S_1	2	S_2
S_1	3	S_3
S_2	0	S_2
S_2	1	S_2
S_2	2	S_2
S_2	3	S_3
S_3	0	S_3
S_3	1	S_3
S_3	2	S_3
S_3	3	S_3